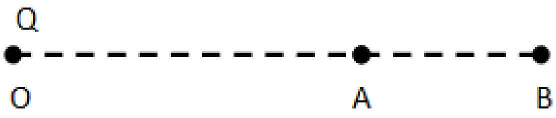
Q1.A point charge Q is placed at point O as shown in the figure, is the potential difference $V_A - V_B$ positive, negative or zero, if Q is (i) positive (ii) negative?

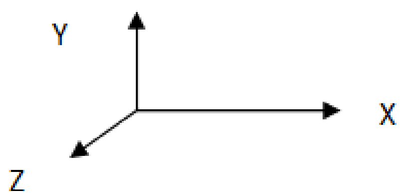


Answer: We know that potential at point due to a point charge V is given by

$V = \frac{Q}{4\pi\varepsilon_0 r}$	$V_A = \frac{Q}{4\pi\varepsilon_0 (OA)}$	$V_B = \frac{Q}{4\pi\varepsilon_0 (OB)}$
When Q is positive , OA < OB	$V_A > V_B$	(i) $V_A - V_B$ is positive.
When Q is Negative, OA < OB	$V_A < V_B$	(ii) $V_A - V_B$ is Negative

Q2.A plane electromagnetic wave travels in vacuum in z – direction. What can you say about the direction of electric and magnetic field vectors?

Answer: We know that the direction of propagation of electromagnetic wave is perpendicular to the plane containing electric vector $\vec{E}$ and magnetic vector $\vec{B}$. Considering the right handed system of axis



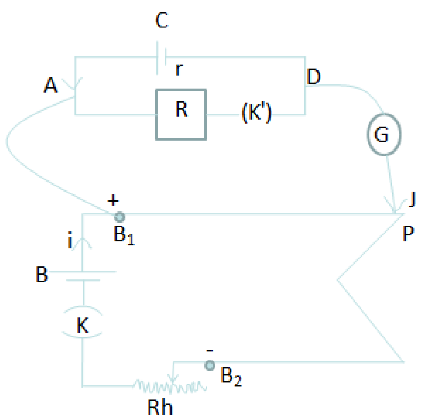
Direction of propagation is given by $\vec{E}\ X\ \vec{B}$.

Therefore if direction of propagation is along Z axis then $\vec{E}$ is along X-axis, $\vec{B}$ is along Y-axis.

Q3.A resistance R is connected across a cell of emf ε and internal resistance r. A potentiometer now measures the potential difference between the terminals of the cell as V. Write the expression for 'r' in terms of ε , V and R.

Answer: $r = \frac{R(l_1 - l_2)}{l_2}$

[Please note to answer above question in examination you can directly write above expression but you must know actual how experiment is performed to find internal resistance using potentiometer.



1. Key K′ is kept open no current flows through R i.e. The circuit of the test cell is an open circuit. The balance point is found by using the test cell C.

   The emf of the test cell $E = i\rho l_1 \rightarrow (1)$

2. The Key K′ is closed a current flows through the circuit of the test cell producing a potential difference across AD

   $V = IR = \left(\frac{E}{R+r}\right)R \rightarrow (2)$

   Using the potential difference the balance point is found

   $V = i\rho l_2 \rightarrow (3)$

   Dividing equation (1) by equation (3)

   $\frac{E}{V} = \frac{i\rho l_1}{i\rho l_2} = \frac{l_1}{l_2}$

   $\frac{E}{\left(\frac{E}{R+r}\right)R} = \frac{l_1}{l_2}$

   $r = \frac{R(l_1 - l_2)}{l_2}$

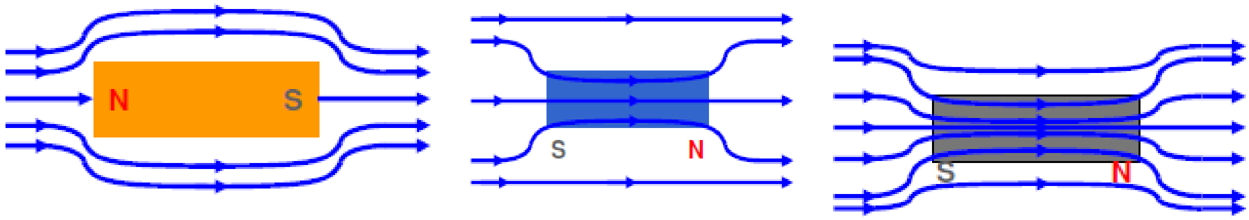
]

Q4. The permeability of magnetic material is 0.9983. Name the type of magnetic materials it represents.

Answer: Here given permeability 0.9983 (nearly equal to 1) hence magnetic material is Paramagnetic.

[You must know permeability is degree of allowance given by material to be passage of lines of force

Diamagnetic	Paramagnetic	Ferro magnetic
$\mu_r << 1$	μ_r ~1 (nearly equal to 1)	$\mu_r >>1$



Q5.Repeate question

Q6. In a transistor, doping level in base is increased slightly. How will it affect (I) collector current and (II) base current?

Answer: We know that the collector current is approximately beta (β) times the base current.

(i) Collector current will decrease
(ii) Base Current will increase

Q7. Define the term 'watt less current'.

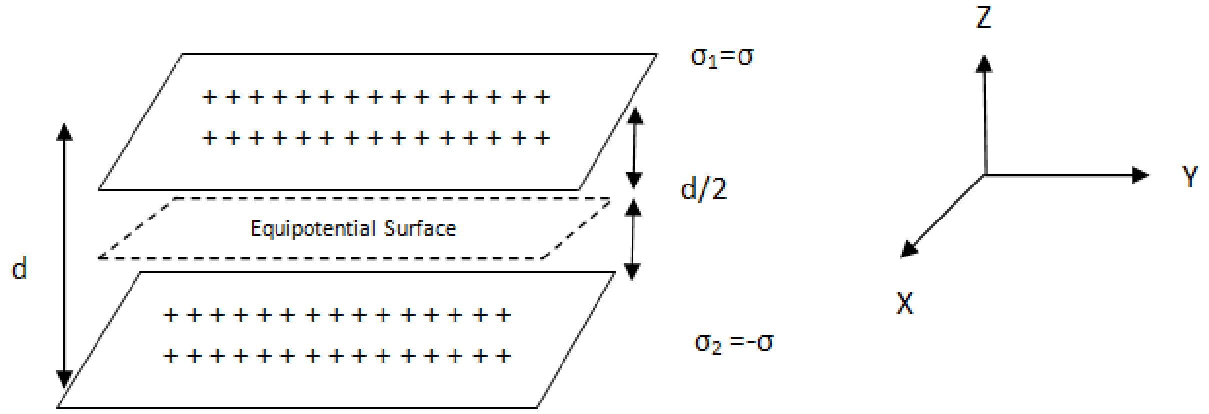
Answer: If the resistance offered by circuit is zero then the power dissipation is also zero such current does not perform any work. The component of current which does not contribute to the average power dissipation is called wattless current

Q8. When monochromatic light travels from one medium to another its wavelength changes but frequency remains the same. Explain.

Answer: velocity = frequency x wavelength, since velocity changes while passing through medium either frequency must change of wavelength must change. Nature requires that the phase of an electromagnetic wave must remain continuous across a boundary. In order for the phase to be continuous for all time the frequency must be constant, hence only wavelength changes.

Q9. Two uniformly large parallel thin plates having charge densities $+\sigma$ and $-\sigma$ are kept in the X – Z plane at a distance'd' apart. Sketch an equipotential surface due to electric field between the plates. If a particle of mass m and charge '-q' remains stationery between the plates, what is the magnitude and direction of this field?

Answer: Equipotential surface is shown by dotted line (potential V=0).



Magnitude of the field

From the figure we find that the net electric field in the middle region where charge –q kept is:

$$E = E_1 - E_2$$

$$= \left(\frac{\sigma_1}{2\epsilon_0} - \frac{\sigma_2}{2\epsilon_0}\right) = \frac{1}{2\epsilon_0}(\sigma_1 - \sigma_2) = \frac{1}{2\epsilon_0}(\sigma + \sigma) = \frac{\sigma}{\epsilon_0}$$

The force due to mass that is gravitation is balanced by force due to electric field

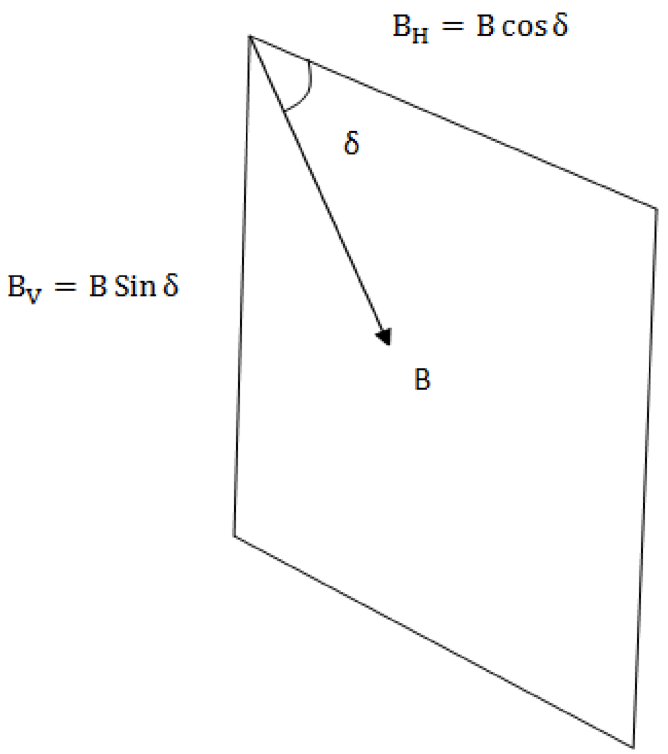
mg = qE or E= mg/q

In this case, the electric field intensity is constant between the sheets. In other words, we get a uniform electric field between the plates.

Direction: Plane of the equipotential surface lies in X-Y plane, since gravity is balance by electric force hence net force due to electric field must be upward that is along Z – axis, as per right handed system of axis shown.

Q10. A magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian has its north tip down at 60^0 with the horizontal. The horizontal component of earth's magnetic field at the place is known to be 0.4 G. Determine the magnitude of the earth's magnetic field at the place.

Answer:



Earth's magnetic field B is divided into two components, horizontal B_H and vertical B_V.

Given :

$\delta = 60^\circ, B_H = 0.4\ G, B = ?$

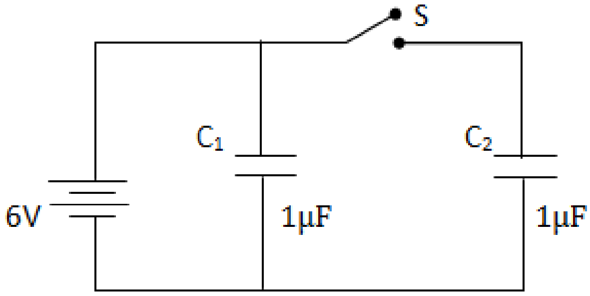
$$B_H = B\cos\delta$$

$$B = \frac{B_H}{\cos\delta} = \frac{0.4}{\cos 60^\circ}$$

$$= \frac{0.4}{1/2} = 0.4 \times 2$$

$$B = 0.8\ T.$$

Q11. Figure shows two identical capacitors, C_1 and C_2, each of 1 μF capacitance connected to a battery of 6 V. Initially switch 'S' is closed. After sometime 'S' is left open and dielectric slabs of constant K=3 are inserted to fill completely the space between the plates of the two capacitors. How will the (I) charge and (II) potential difference between the plates of the capacitors be affected after the slabs are inserted?



Answer:

When S is closed	When S is opened
Charge on C_1 , $q_1=C_1V$ =1x6 =6 μC Charge on C_2 , $q_2=C_2V$=1x6=6 μC Both the capacitors have same charge 6 μC and same potential 6 Volt.	Since Dielectric constant 3 is inserted on both capacitors. Capacitance of C_1 increased to C_1' $=3C_1=3\mu F$ Capacitance of C_2 increased to C_2' $=3C_2=3\mu F$ Potential =6V therefore Charge on C_1 $=q_1'=CV=3X6=18\mu C$ As C_2 is disconnected, charge on $C_2=q_2'=6$ μC Capacitance of $C_2= C_2'$=KX1=3 μF Potential at C_2=Charge/Capacitance=6/3=2 Volt.

Q12. Two convex lenses of same focal length but of aperture A_1 and A_2 ($A_1 < A_2$), are used as the objective lenses in two astronomical telescopes having identical eyepieces. What is the ratio of their resolving power? Which telescope will you prefer and why? Give reasons.

Answer: We know that resolving power is proportional to the aperture and is given by $\mathrm{R.P.} = \frac{a}{1.22\lambda}$

$$\therefore \frac{(\text{Resolving Power of Lens})_1}{(\text{Resolving Power of Lens})_2} = \frac{A_1}{A_2}$$

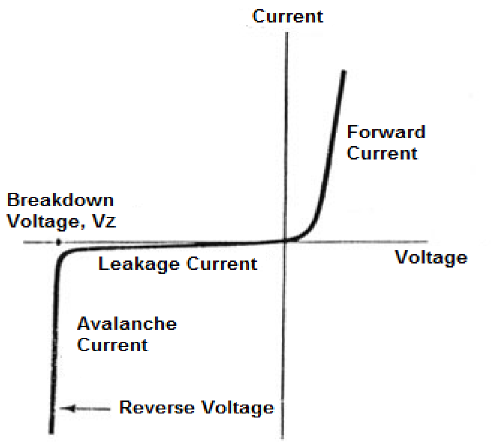
The telescope with objective of aperture A_2, should be preferred for viewing as this would

i. Give a better resolution.
ii. Have a higher light gathering power of telescope.

Q13. Repeat question

Q14. Name the semiconductor device that can be used to regulate an unregulated dc power supply. With the help of I – V characteristics of this device, explain its working principle.

Answer: Semiconductor device used to regulate unregulated dc power supply **is** Zener diode.



Working Principle:

V-I characteristics of zener diode indicates in forward bias region Zener diode is identical to that of Normal PN junction diode. In reverse bias zener diode acts as an open circuit and only small current in order of (1 μA) flows through the diode as shown in the figure. If the applied reverse voltage exceeds the breakdown voltage then break down occurs. In breakdown region zener diode provides almost constant voltage and the current through the zener increases rapidly.

A regulator is used to provide constant output voltage to the load connected in parallel with the regulator. Zener diode regulates the output voltage until the zener current falls below the minimum current in breakdown region.

Zener diodes are used to stabilize or regulate the voltage source against load or supply variations as it provides constant voltage in the reverse bias even though input voltage is varying after the breakdown.

Q15. How are infra red waves produced? Why are these referred to as 'heat waves'? Write their one important use.

Answer: Infra red waves are produced by

(1) Hot bodies and molecules.

(2) Laser

(3) In Globar as thermal light source where a Silicon carbide rod when electrically heated by to 1000 to 1650 ^{0}C emits infra red waves

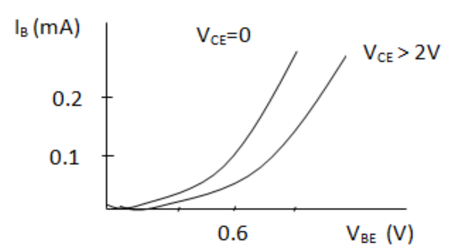
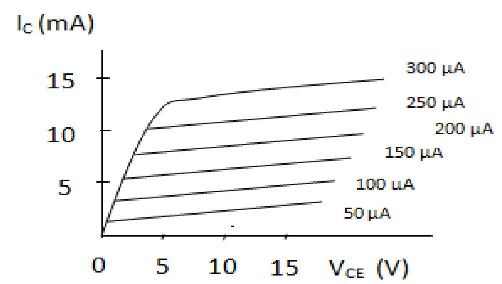
(4) Radiation from Sun

Since most of the thermal radiation emitted by objects near room temperature are Infra red hence referred as Heat Waves.

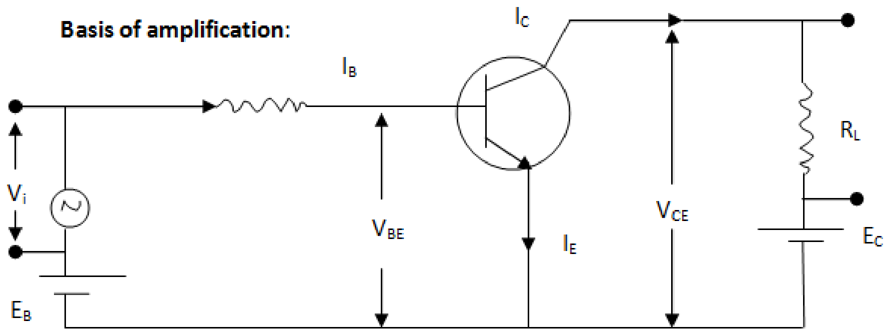
Use: In night-vision devices infrared illumination are used to see people or animals secretly.

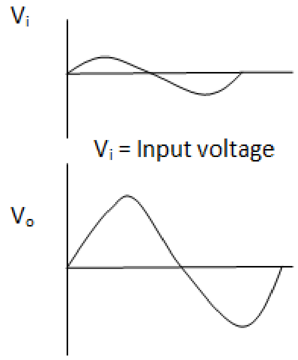
Q16. Draw the transfer characteristic curve of a base biased transistor in CE configuration. Explain clearly how the active region of the V_o versus V_i curve in a transistor is used as an amplifier.

Answer:

 Input Characteristics	The input characteristics like forward biased p-n junction. If the biasing voltage is small as compared to the height of the potential barrier at the junction, the current I_B is very small. When voltage is more than the barrier height, the current rapidly increases. Here most of the electrons diffused across the junction go to the collector, the net base current is very small or the order of microamperes even at large values of V_{BE}
 Output Characteristics	In the output characteristics, for small values of collector voltage, the collector base junction is reverse biased as the base is more at positive potential. The current Ic is small. As the electrons are forced from emitter side, the collector current Ic is still large as compared to reverse biased p-n junction. As voltage Vc is increased, the current rapidly increases and becomes constant once the junction is forward biased. When base current increases ,collector current is high and increases rapidly in forward bias.

Basis of amplification:





Above circuit shows n-p-n transistor in common emitter mode. The battery E_B provides biasing voltage for base-emitter junction. V_{CE} voltage between collector and emitter maintained by using battery E_C.
The base emitter junction is forward biased and so the electrons of the emitter flow towards the base. As the base region is very thin (of the order of micrometer) and collector is also maintained at a positive potential, most of the electrons cross the base region and move into the collector.
The holes in the base region may diffuse into the emitter due to the forward biasing of the base-emitter junction. The electrons coming from the emitter may recombine with some of the holes in the base. If the holes are lost in this way, the base will become negatively charged and will obstruct the incoming electrons from the emitter. If the base current IB is increased by a small amount, the effect of the hole-diffusion and hole-electron recombination may be neutralised and the collector current will be increased. Thus a small change in the current IB in the base circuit controls the larger current Ic in the collector circuit. This is the **basis of amplification in transistor.**

The input signal to be amplified is connected in series with the biasing battery E_B in the circuit. A load resistor having a large resistance R_L is connected in the collector circuit, the output voltage is taken across resistor. As V_{BE} (base emitter voltage) changed, base current I_B changes, which results changes in collector current I_C, current gain $\beta = \Delta I_C / \Delta I_B$, voltage across $R_L = \Delta V = R_L \Delta I_C$

Q17. I. Define modulation index.
II. Why is the amplitude of modulating signal kept less than the amplitude of carrier wave?

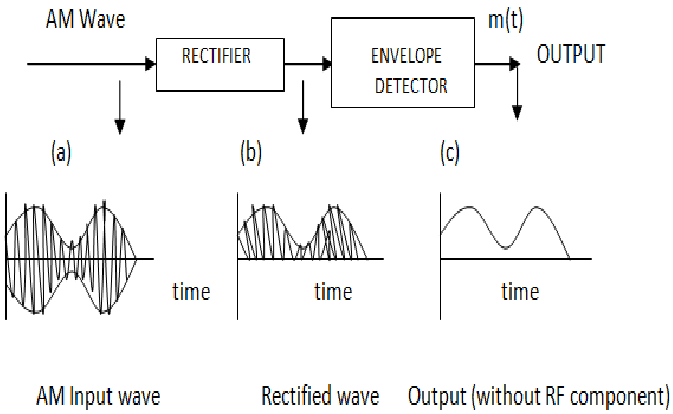
Answer:

I. **Modulation index** is defined as ratio of amplitude of the modulated wave to the amplitude of the carrier wave.

$$\mu = \frac{A_m}{A_c}$$

Where A_m = Amplitude of the modulated wave
A_c = Amplitude of the carrier wave.

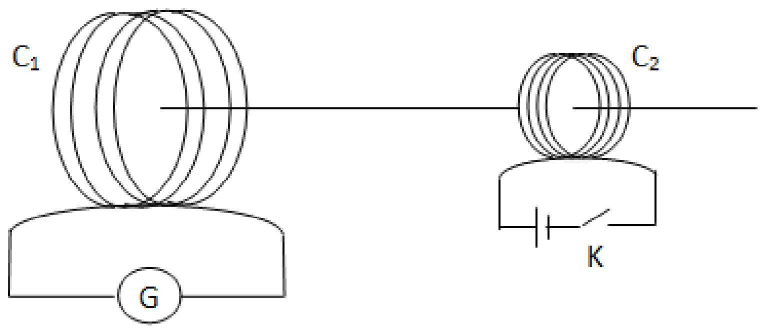
Thus the degree, to which the carrier wave is modulated, is called modulation index.



Modulating signal uses peak value of carrier, the envelop of the modulating signal varies above and below the peak carrier amplitude. If the amplitude of the modulating signal is greater than amplitude of the carrier then **distortion** will occur, this will cause incorrect information transmission. Therefore peak value of the modulating signal should be less than peak value of carrier.

Q18. A current is induced in coil C_1 due to the motion of current carrying coil C_2.

a. Write any two ways by which a large deflection can be obtained in the galvanometer G.
b. Suggest an alternative device to demonstrate the induced current in place of a galvanometer.



Answer:

(a) For deflection in the galvanometer electromagnetic induction should happen, i.e. magnetic lines of force should cut plane of C_1 and magnetic lines (flux) should also vary.

1. When the tapping key K is released, current in C_2 reduces to zero, magnetic flux cutting C_1 will change hence the galvanometer shows momentary deflection, the pointer in the galvanometer returns to zero almost instantly. Similarly if the K is pressed, current in C_2 increases from 0 hence magnetic flux cutting C_1 will change, we can observe deflection in C_1.If we varying this current in C_2 by using a Rheostat in the circuit of C_1 we can get large deflection in C_1 or moving coil C_2 faster.

2. For large deflection we can insert one ferro-magnetic rod say iron rod along the axis of C_1 and C_2, this helps to increase permittivity of the medium and negligible loss of magnetic lines which originates at the centre of C_2 and cuts plane of C_1 .

(b) Alternative device to detect em induction, In place of C_1 coil we can keep small magnetic needle to detect magnetic field produced by C_2.

Q19. (a) Define the terms (I) drift velocity, (II) relaxation time.

(b) A conductor of length L is connected to a dc source of emf ε. If this conductor is replaced by another conductor of same material and same area of cross section but of length 3L, how will the drift velocity change?

Answer:

a. (I) **Drift velocity:** The average velocity with which the free electrons get drifted towards the positive terminal under the effect of the applied external electric field.

(II) **Relaxation time:** The average time interval between two successive collisions of an electron with the ions or atoms of the conductor. Electron relaxes during the time interval τ is of the order of $10^{-14}\ s$.

b. We know that drift velocity is inversely proportional to length $V_d \ \alpha \ \frac{1}{l}$

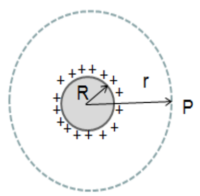
Therefore $V'_d \ \alpha \ \frac{1}{3l}$ or $V'_d = V_d/3$, therefore drift velocity will be one-third.

Q20. Using Gauss's Law obtain the expression for the electric field due to a uniformly charged thin spherical shell of radius R at a point outside the shell. Draw a graph showing the variation of electric field with r, for r > R and r < R.

Answer: Q= charge on the surface of the shell
R=Radius of the shell
r=OP=distance of the given point from the centre of the shell.
ε=Permittivity of the medium
Let E= Electric intensity at P due to the charged shell.

Let us imagine a concentric shell of radius r passing through P this is our Gaussian surface.
Surface area of the Gaussian surface A= $4\pi r^2$
Since the lines of force leave the surface of a charged body perpendicularly hence the lines of force are all radial to the charged sphere and hence force are all radial to the charged sphere and hence they are also radial to the Gaussian surface and cutting the surface perpendicularly.

Case I :When Point P lies outside the charged shell



Using definition of flux through the Gaussian surface
$\phi = EA = E.4\pi r^2$ (1)
Applying the Gauss's theorem the flux through the Gaussian surface

$$\phi = \frac{\text{Charged enclosed}}{\varepsilon} = \frac{Q}{\varepsilon} \rightarrow (2)$$

Equating (1) and (2):

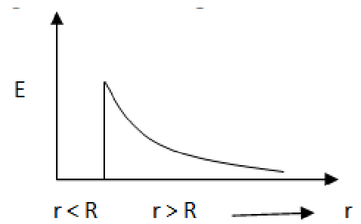
$$E.4\pi r^2 = \frac{Q}{\varepsilon}$$

$$E = \frac{1}{4\pi\varepsilon}\frac{Q}{r^2} \rightarrow (3)$$

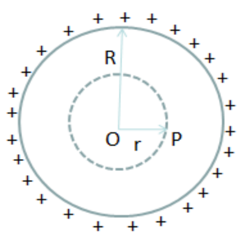
If we imagine a point charge Q at the center of the shell intensity at P

$$E = \frac{1}{4\pi\varepsilon}\frac{Q}{r^2} \rightarrow (4)$$

Graph:



Case II :When Point P lies inside the charged shell



$$\phi = \frac{\text{Charged enclosed}}{\varepsilon} = \frac{0}{\varepsilon} = 0 \rightarrow (5)$$

From equation (1) and (5):

$$E.4\pi r^2 = 0$$

$$\because r \neq 0 \therefore E = 0$$

Intensity at a point inside a charged shell is zero.

Q21. An electron and a photon each have a wavelength 1.00 nm. Find

i. Their momentum.
ii. The energy of the photon and
iii. The kinetic energy of electron.

Answer: Given $\lambda = 10^{-9}\mathrm{m}$

i. For electron or photon's momentum,

$$p = \frac{h}{\lambda} = \frac{6.63\times10^{-34}}{10^{-9}} = 6.63 \times 10^{-25}\ \mathrm{kg\ m/s}$$

ii. $E = \frac{hc}{\lambda} = \frac{(6.63\times10^{-34})\times(3\times10^{8})}{10^{-9}\times(1.6\times10^{-19})} = 1243$ eV [1 eV= 1.6×10^{-19} Joule]

iii. $E_k = \frac{1}{2}\frac{p^2}{m}$

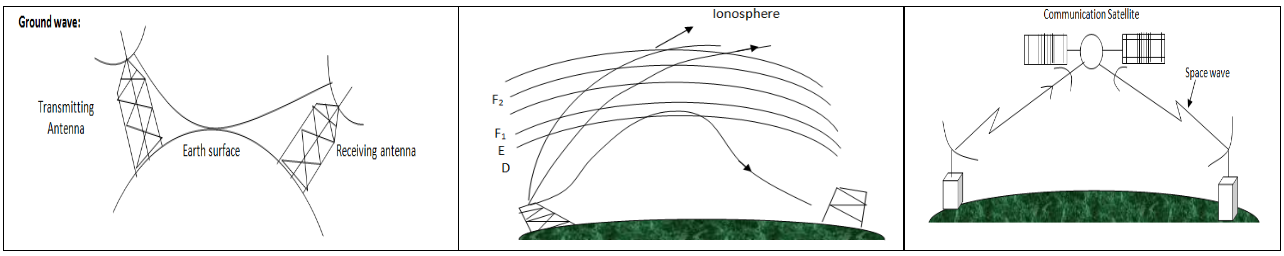
$$= \frac{1}{2}\frac{(6.63\times10^{-25})^2}{9\times10^{-31}\times(1.6\times10^{-19})}$$

$= 1.52$ eV. [1 eV= 1.6×10^{-19} Joule]

Q22. Draw a schematic diagram showing the (I) ground wave (II) sky wave and (III) space wave propagation modes for Electromagnetic waves. Write the frequency range for each of the following:

i. Standard AM broadcast
ii. Television
iii. Satellite communication

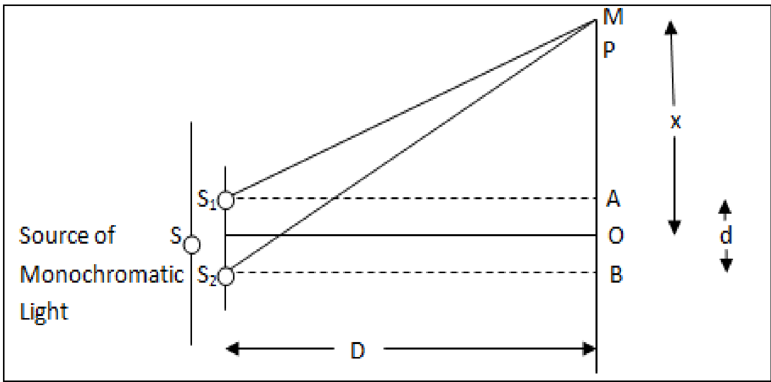
Answer:



i. Standard AM broadcast – 540 – 1600 kHz
ii. Television 54-890 MHz
iii. Satellite communication 5.925 – 6.425 GHz Uplink, 3.7-4.2 GHz Downlink

Q23. Describe Young's double slit experiment to produce interference pattern due to a monochromatic source of light. Deduce the expression for the fringe width.

Answer:



Light waves coming out from S incident on both slit S_1 and S_2. The spherical waves from S_1 and S_2 being coherent source of light will produce interference fringes on the screen M.

In right angle Δ S_1 AP, we have

$(S_1P)^2 = (S_1A)^2 + (AP)^2$

$S_1P = \sqrt{D^2 + (x - \frac{d}{2})^2} = \sqrt{D^2[1 + \frac{(x-\frac{d}{2})^2}{D^2}]}$

$S_1P = \mathrm{D}\left[1 + \frac{(x-\frac{d}{2})^2}{D^2}\right]^{\frac{1}{2}}$

By Binomial Theorem and neglecting higher terms, we have

$S_1P = \mathrm{D}\left[1 + \frac{1}{2}\frac{(x-\frac{d}{2})^2}{D^2}\right] = \mathrm{D} + \frac{(x-\frac{d}{2})^2}{2D}$

Similarly $S_2P = \mathrm{D} + \frac{(x+\frac{d}{2})^2}{2D}$

Hence path difference = S_2 P – S_1 P

$= \mathrm{D} + \frac{(x+\frac{d}{2})^2}{2D} - \mathrm{D} - \frac{(x-\frac{d}{2})^2}{2D}$

$= \frac{1}{2\mathrm{D}}\left[x^2 + \frac{d^2}{4} + xd - x^2 - \frac{d^2}{4} + xd\right]$

$= \frac{1}{2\mathrm{D}} . 2\mathrm{xd} = \frac{xd}{D}$

Now the intensity at point P is maximum or minimum according as the path difference is an integral multiple of wavelength or an odd integral multiple of half wavelength.

(i) **For bright fringe (constructive interference):** We will have constructive interference resulting in a bright fringe when path difference = nλ

$\frac{xd}{D} = \text{n}\lambda$ => x = $\frac{\text{n}\lambda D}{d}$

$\therefore\ x_n = \frac{\text{n}\lambda D}{d}$, where n=0, $\pm$ 1, $\pm$2, … …

Since the separation between the centres of two consecutive bright fringes is called **fringe width**. It is denoted by β.

$\therefore\ \beta = x_{n+1} - x_n$

$or, \beta = \frac{(n+1)\lambda\ \text{D}}{d} - \frac{n\lambda\ \text{D}}{d} = \frac{\lambda\ \text{D}}{d}$ (n+1-n)

$\therefore\ \beta = \frac{\lambda\ \text{D}}{d}$

(ii) **For dark fringe (destructive interference):** We will have destructive interference resulting in a dark fringe when path difference = (2n + 1) $\frac{\lambda}{2}$

$\frac{xd}{D}$ = (2n + 1) $\frac{\lambda}{2}$ => x = $\frac{(2n+1)\lambda\ \text{D}}{2d}$

$\therefore\ x_n = \frac{(2n+1)\lambda\ \text{D}}{2d}$, where n=0, $\pm$ 1, $\pm$2, … …

Fringe width, $\beta = x_{n+1} - x_n$

= $\frac{[2(n+1)+1]\lambda\ \text{D}}{2d} - \frac{(2n+1)\lambda\ \text{D}}{2d}$

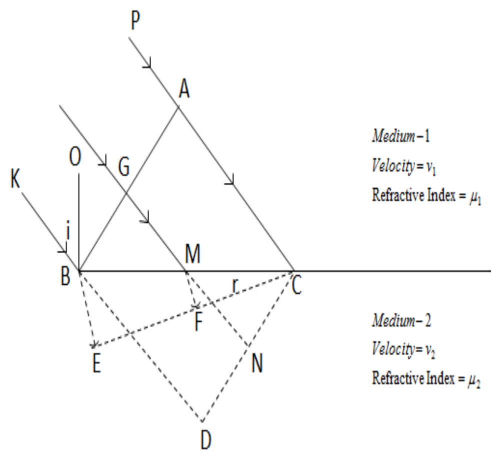
= (2n+2+1-2n-1) $\frac{\lambda\ \text{D}}{2d} = \frac{2\lambda\ \text{D}}{2d}$

$\therefore\ \beta = \frac{\lambda\ \text{D}}{d}$

Hence all bright and dark fringes are of equal width.

Or

Use Huygens's principle to verify the laws of refraction



BC is the section of the refracting surface by the plane of the paper i.e. the out face, separating two media of refractive indices μ_1 and μ_2 ($\mu_1 < \mu_2$)
AB is the section of the incident plane wave front by the plane of the paper at an
instant of time t=0
Both the refracting surface and the plane wave front are perpendicular to the plane of the paper.

v_1 & v_2 = Velocity of light in medium-1 and medium-2 respectively.

PA and KB are perpendiculars dropped on the incident wave front AB and hence they are the incident rays. BO is normal drawn at B on the refracting surface.

$$\angle KBO = i = \text{angle of incidence}$$
$$\angle KBA = 90 \quad \therefore \angle OBA = 90 - i$$
$$\angle OBC = 90 \quad \therefore \angle ABC = 90 - (90 - i) = \text{i}$$

Hence angle of incidence can also be defined as the angle between the incident wave front

and the refracting surface. Similarly the angle of refraction can also be defined as the angle between the refracted wave front and the refracting surface.

Had the two medium been same, then the rays from the point G & B would have gone up to N & D respectively during the time t in which ray from a goes up to C so that

$$AC = GN = BD = v_1 t \qquad (1)$$

Hence CND would have been the position of the wave front after time t. But since the two medium are different let the ray from B go up to E in time t. So that

$$t = \frac{AC}{v_1} = \frac{BE}{v_2} \rightarrow (2)$$

With B as center and BE as radius we imagine a sphere the section of which by the plane of the paper gives an arc of a circle. CE is drawn tangential to that rc. Then a plane through CE and perpendicular to the plane of the paper would represent the refracted wave front after time t provided we can show that ray from any other point G on the incident wave front also touches the plane through CE after time t.To prove it, from M, MF is drawn perpendicular to CE.

ΔBDC & ΔMNC are similar $\therefore \frac{BD}{MN} = \frac{BC}{MC} \rightarrow (3)$

ΔBEC & ΔMFC are similar $\therefore \frac{BE}{MF} = \frac{BC}{MC} \rightarrow (4)$

From (3) & (4) $\frac{BD}{MN} = \frac{BE}{MF}$

Putting equation (1) & (2) $\frac{v_1 t}{MN} = \frac{v_2 t}{MF}$ or $\frac{MF}{v_2} = \frac{MN}{v_1}$

That is time taken to go from M to F in medium 2 and time taken to go from M to N in medium 1 is same.

Hence the time taken by the light to go from G to M + M to N is same as the time taken to go from G to M + M to F i.e. from G to F.

Hence CE is the refracted wave front after time t.

$\therefore \angle BCE$ = angle between the refracting surface and the refracted wave front = angle of refraction = r (using definition)

We now prove Snell's law :

From ΔABC : $\sin i = \frac{AC}{BC}$

From ΔABC : $\sin r = \frac{BE}{BC}$

$$\therefore \frac{\sin i}{\sin r} = \frac{AC/BC}{BE/BC} = \frac{AC}{BE}$$

Putting equation (1) and (2): $\frac{\sin i}{\sin r} = \frac{v_1 t}{v_2 t}$

$$\frac{\sin i}{v_1} = \frac{\sin r}{v_2}$$

Multiplying both sides by v_0 where v_0 is the velocity of light in vacuum.

$$\frac{v_0}{v_1}\sin i = \frac{v_0}{v_2}\sin r$$

$\mu_1 \sin i = \mu_2 \sin r$ First law proved

From the nature of construction it is evident that the incident ray refracted ray & normal at the point of incidence all lay on the same plane i.e. on the plane of the paper.

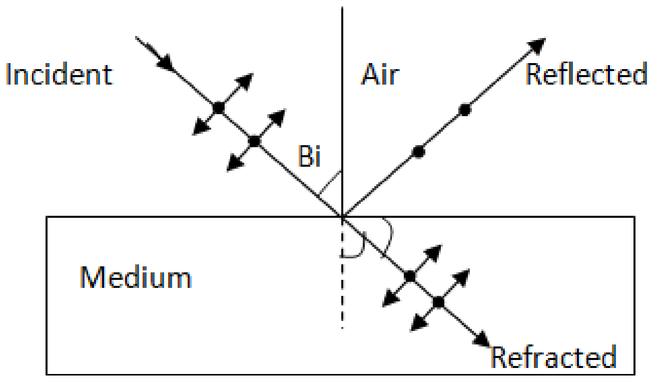
Q24.

a. Describe briefly, with the help of suitable diagram, how the transverse nature of light can be demonstrated by the phenomenon of polarization.

b. When unpolarized light passes from air to a transparent medium, under what condition does the reflected light get polarized?

Answer:

i. Transverse nature of light through Polarization**:**



The independent light waves whose planes of vibrations are randomly oriented about the direction of propagation are said to be unpolarized light. When unpolarized light is incident on the boundary between two transparent media, the reflected light is polarized with electric vector perpendicular to the plane of incidence when the refracted and reflected rays make a right angle with each other. Whenever unpolarized light is incident from air to a transparent medium at an angle of incidence equal to polarizing angle, the reflected light gets polarized as shown in the diagram above. Thus Light can be polarized by reflecting it from a transparent medium. The extend of polarization depend on the angle of incidence, called Brewster's angle.

ii. Whenever the reflected and refracted rays are perpendicular to each other.

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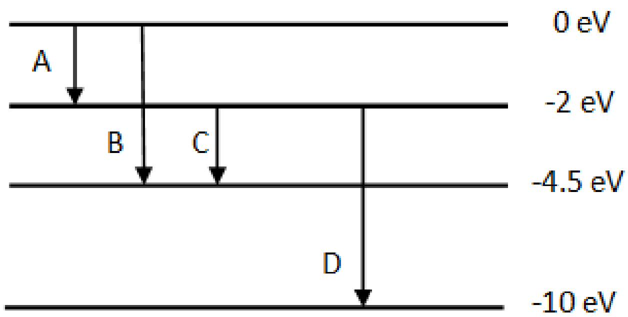
Q25. The energy levels of a hypothetical atom are shown below. Which of the shown transitions will result in the emission of a photon of wavelength 275 nm?
Which of these transitions correspond to emission of radiation of (I) maximum and (II) minimum wavelength?



We know that energy emitted E = hc/λ

Given: λ = 275nm=275X10^{-9} m

$$E = \frac{6.63\times10^{-34}x3x10^{8}}{275X10^{-9}x1.6X10^{-19}} = 4.5eV$$

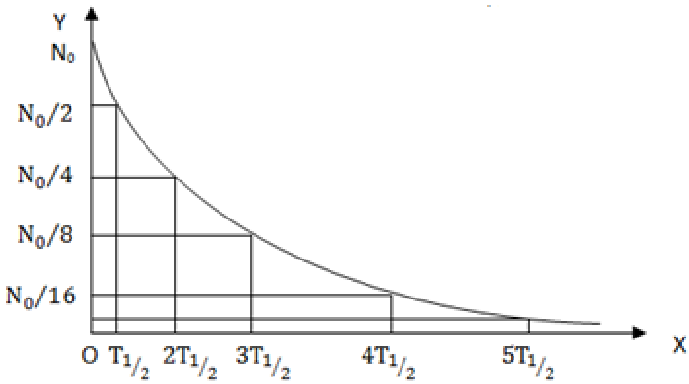
I. Element A has corresponds to maximum wavelength (621 nm).
II. Element D has corresponds to minimum wavelength (155 nm).

Q26. State the law of radioactive decay.
Plot a graph showing the number (N) of undecayed nuclei as a function of time (t) for a given radioactive sample having half life $T_{1/2}$.
Depict in the plot the number of undecayed nuclei at (i) $t = 3\ T_{1/2}$ and (ii) $t = 5\ T_{1/2}$.

Answer: Radioactive decay law**:** The number of atoms disintegrated per second at any instant is directly proportional to the number of radioactive atoms actually present at that time.

$$-\frac{dN}{dt} \propto N \text{ or } -\frac{dN}{dt} = \lambda N$$

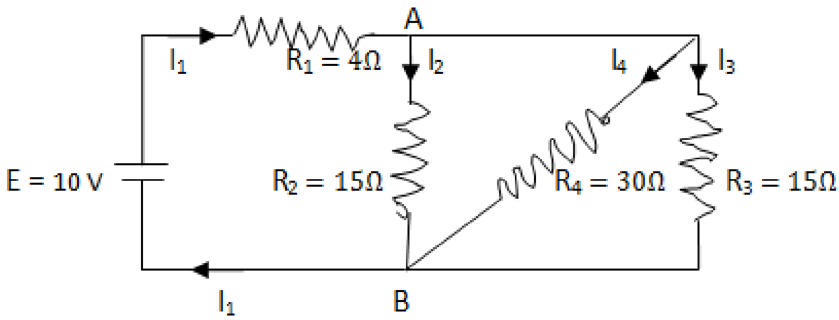
Graph:



Taking number of atoms along Y axis and time along X axis following we find following trend plotted.

The un-decayed nuclei at (i) $t = 3\ T_{1/2}$ and (ii) $t = 5\ T_{1/2}$ marked

Q27. In the circuit shown, $R_1 = 4\ \Omega,\ R_2 = R_3 = 15\ \Omega,\ R_4 = 30\ \Omega$ and $E = 10$ V. Calculate the equivalent resistance of the circuit and the current in each resistor.



Answer: R_2, R_3 and R_4 are in parallel combination.

$$\frac{1}{R_{234}} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} = \frac{1}{15} + \frac{1}{30} + \frac{1}{15} = \frac{1}{6}$$

$R_{234} = 6\ \Omega$

R_{234} and R_1 are in Series

$R_{1234} = R_{234} + R_1 = 6 + 4$

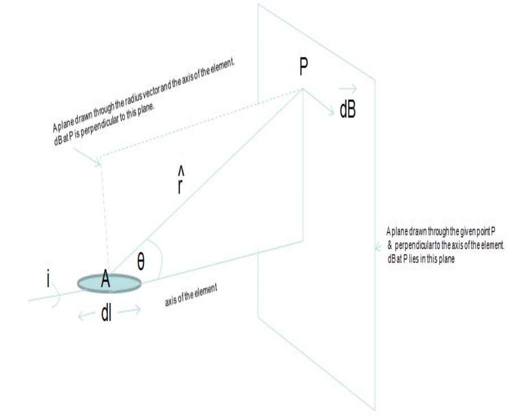
$R_{1234} = 10\ \Omega$

$$I = \frac{E}{R_{1234}} = \frac{10}{10} = 1\text{ A}$$

Q28. State Biot-Savart law, giving the mathematical expression for it.
Use this law to derive the expression for the magnetic field due to a straight conductor carrying current at a point along its axis.

How does a circular loop carrying current behave as a magnet?

Answer: Statement of Biot Savart Law: The magnitude of the magnetic field due to current element is directly proportional to the current, the element length and inversely proportional to the square of the distance of the field point. The direction of magnetic field is perpendicular to the plane containing current element and distance.



Given

i = current flowing through the element

dl = the length of the element of the conductor

r = AP = the distance of the given point from the element of the conductor

$\hat{r}$ = a unit vector along AP

$\hat{dl}$ = a vector of unit length along the axis of the element in the direction of current

θ = the angle between $\hat{r}$ and $\hat{dl}$

Let $\vec{db}$ = magnetic induction vector at the point P due to the current element

Experimentally it has been found that

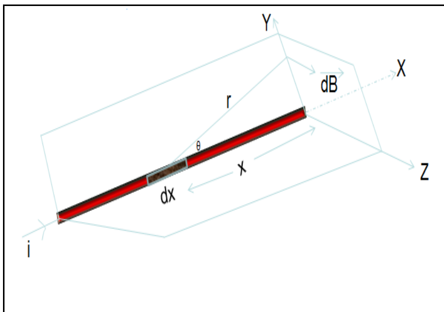
$|\vec{db}| \propto i, |\vec{db}| \propto dl, |\vec{db}| \propto \frac{1}{r^2}, |\vec{db}| \propto \sin\theta$

$db \propto \frac{idl\sin\theta}{r^2}$

$$\text{db} = \frac{\mu_0}{4\pi}\frac{i}{r^2}\left(\vec{dl} \times \hat{r}\right)$$

$$\text{db} = \frac{\mu_0}{4\pi}\frac{i}{r^3}\left(\vec{dl} \times \vec{r}\right)$$

To find the magnetic field at a point due to an infinitely long straight conductor carrying current.



Let AP = r, $O\hat{A}P = \theta$

Applying Biots law the magnetic induction vector at P due to the current element at A

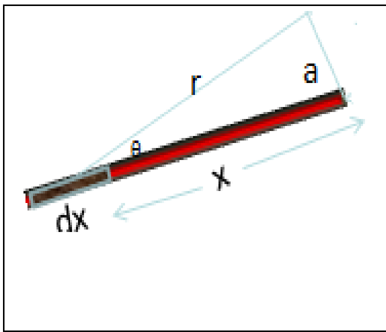
$$\vec{dB} = \frac{\mu_0}{4\pi}\frac{i}{r^2}\vec{dx} \times \hat{r} \rightarrow (1)$$

The magnitude of this induction vector

$$\text{dB} = \frac{\mu_0}{4\pi}\frac{i}{r^2}dxSin\theta \rightarrow (2)$$

$$B = \int dB = \int \frac{\mu_0}{4\pi}\frac{i}{r^2}dxSin\theta \rightarrow (3)$$

Equation (3) contains three variables x, r & θ hence it is represented in terms of one variable only.



$$\text{cosec}\theta = \frac{r}{a}, r = a\text{cosec}\theta$$

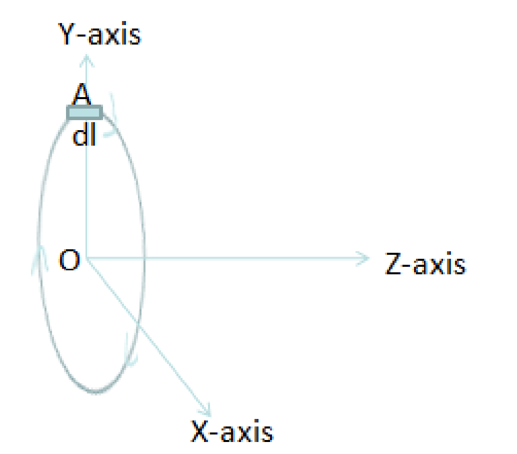
When $x = -\infty, \theta = 0$ and when $x = +\infty, \theta = 180^0 = \pi$

$$\therefore B = \frac{\mu_0}{4\pi} i \int_0^{\pi} \frac{a\cos ec^2\theta.\sin\theta}{a^2\cos ec^2\theta} d\theta$$

$$B = \frac{\mu_0}{4\pi}\frac{i}{a}\int_0^{\pi}\sin\theta d\theta = \frac{\mu_0}{4\pi}\frac{i}{a}[-\cos\theta]_0^{\pi} = \frac{\mu_0}{4\pi}\frac{2i}{a} \rightarrow (5)$$

Equation(5) is known as Biot-Savart formula and gives the magnitude of the magnetic field at a point near a straight conductor carrying current. The direction of the magnetic field which was found to be along Z axis in the figure can be stated as follows:

"Grasp the conductor in right hand with the thumb in the direction of current. The curl of the finger tips gives the direction of the magnetic field."



The direction of magnetic field is along the axis of the coil which can be found as follows "Curl the finger of right hand in the direction of current in the coil the thumb gives the direction of the magnetic field".

Thus magnetic flux emitting for this magnetic field produced is along X axis then this side i.e coil facing +X axis will behave as North pole and negative X axis (X' axis) will behave as South pole. Hence coil (loop) behaves as a magnetic dipole.

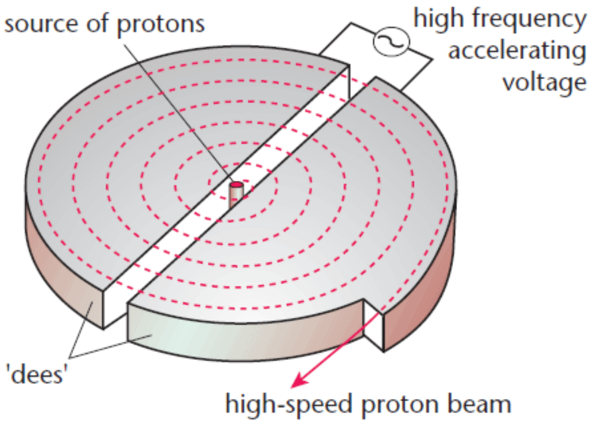
Or,

With the help of labelled diagram, state the underlying principle of a cyclotron. Explain clearly how it works to accelerate the charged particles.

Show that cyclotron frequency is independent of energy of the particle. Is there an upper limit on the energy acquired by the particle? Give reason.

Answer:

Principle: Cyclotrons accelerate charged particle beams using a high frequency alternating voltage which is applied between two "D"-shaped electrodes. An additional static magnetic field B is applied in perpendicular direction to the electrode plane. Thus charge particle acquires very high kinetic energy with the help of relative small electric field.



Working: A cyclotron consists of two D-shaped regions known as dees. In each dee there is a magnetic field perpendicular to the plane of the page. In the gap separating the dees there is a uniform electric field pointing from one dee to the other. When a charge is released from rest in the gap it is accelerated by the electric field and carried into one of the dees. The magnetic field in the Dee causes the charge to follow a half-circle that carries it back to the gap.

While the charge is in the dee the electric field in the gap is reversed, so the charge is once again accelerated across the gap. The cycle continues with the magnetic field in the dees continually bringing the charge back to the gap. Every time the charge crosses the gap it picks up speed. This causes the half-circles in the dees to increase in radius, and eventually the charge emerges from the cyclotron at high speed.

Theory: The centripetal force needed by the charged particle to move in circular track, is provided by the magnetic field.

m= mass of the charge particle, v=velocity, r=radius of the circular track, q=charge, B=Magnetic field

$\frac{mv^2}{r} = qvB$ or $v = \frac{qBr}{m}$, Now period of revolution= $T = \frac{2\pi r}{v} = \frac{2\pi r}{\frac{qBr}{m}} = \frac{2\pi m}{qB}$

Frequency $\nu = \frac{1}{T} = \frac{qB}{2\pi m}$, therefore the frequency of revolution is independent of energy of the particle.

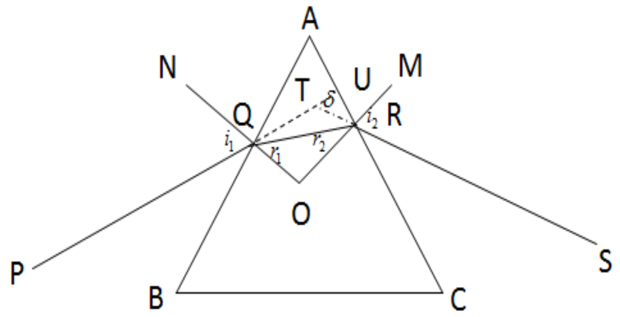
There is an upper limit on the energy acquired by the charged particle. The speed hence energy gain is maximum (at upper limit) when the radius of the moving path is equal to the radius of the dees.

Q29. (a) Draw a ray diagram to show refraction of a ray of monochromatic light passing through a glass prism.

Deduce the expression for the refractive index of a glass in terms of angle of prism and angle of minimum deviation.

(b) Explain briefly how the phenomenon of total internal reflection is used in fibre optics.

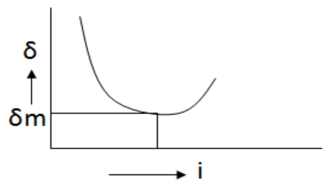
Answer (a): Refraction through prism



A = Angle of prism

Had there been no prism, the ray PQ would have gone straight along PQU, but due to the prism the ray finally goes along RS. Hence the prism has deviated the ray through the angle UTR=δ and is known as angle of deviation.

ON & OM are perpendiculars drawn on AB and AC respectively.



Let

$\angle PQN = i_1 =$ angle of incidence

$\angle RQO = r_1$

$\angle SRM = i_2 =$ angle of emergence

$\angle AQO = \pi/2 = \angle ARO$

In quodrilateral AQOR

Sum of two opposite angles $\angle AQO + \angle ARO = 180^\circ$

Hence AQOR is a co cyclic quadrilateral

$\angle QAR + \angle QOR = 180^\circ$

$A + \angle QOR = 180^\circ \rightarrow (1)$

$In\ \Delta QOR$

$r_1 + r_2 + \angle QOR = 180^\circ \rightarrow (2)$

From equation (1) and equation (2):

$A = r_1 + r_2 \rightarrow (3)$

$\angle TQR = i_1 - r_1$

$\angle TRQ = i_2 - r_2$

$\delta = (i_1 - r_1) + (i_2 - r_2)$

$\delta = (i_1 + i_2) - (r_1 + r_2)$

$\delta = (i_1 + i_2) - A \rightarrow (4)$

From equation (4) we find that the angle of deviation δ depends on the angle of Incidence. Experimentally it is found that as the angle of incidence increases angle of deviation decreases, reaches a minimum value δm for a particular angle of incidence and then increases again.

The minimum value δm called the angle of minimum deviation. It can be analytically proved that when

$i_1 = i_2, r_1 = r_2$ i.e. the path of the ray through the prism is parallel to the base of the prism then only the angle of deviation is minimum.

Condiation for minimum deviation

$i_1 = i_2 \ \& \ r_1 = r_2$

From equation (4)

$$\delta m = i + i - A$$

$$i = \frac{A + \delta m}{2}$$

$$A = r_1 + r_2 = r + r$$

$$A = 2r$$

$$r = \frac{A}{2}$$

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin\left(\frac{A + \delta m}{2}\right)}{\sin \frac{A}{2}}$$

For a thin prism the angle of prism A is very small

$$\sin \frac{A}{2} \approx \frac{A}{2}$$

$$\sin\left(\frac{A + \delta m}{2}\right) \approx \frac{A + \delta m}{2} \approx \frac{A + \delta}{2}$$

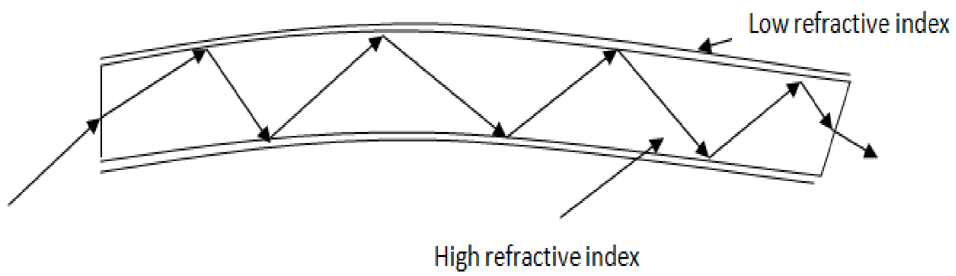
$$\therefore \mu = \frac{\frac{A + \delta}{2}}{\frac{A}{2}}$$

$$or \ A\mu = A + \delta$$

$$\text{or } \delta = A(\mu - 1) \rightarrow (6)$$

Equation (6) gives the deviation produced by thin prism.

(b) Total internal reflection is used in optical fibre: Optical fibre consists of a core and cladding. Refractive index of the material of the core is higher than that of cladding. When a signal, in the form of light, is directed into the optical fibre, at an angle greater than the respective critical angle, it undergoes repeated total internal reflections as shown along the length of the fibre.



Finally the signal comes out from the other end with almost negligible loss of intensity.

OR

(a) Obtain lens makers formula using the expression

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Here the ray of light propagating from a rarer medium of refractive index (n_1) to a denser medium of refractive index (n_2) is incident on the convex side of spherical refracting surface of radius of curvature R.

Answer:

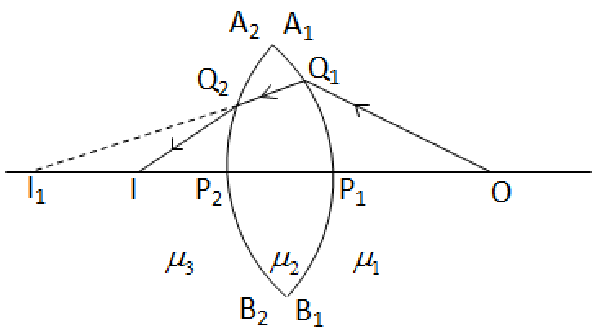


Fig-1

A_1B_1 & A_2B_2 are two spherically curved surfaces separating media of refractive indices μ_2, μ_1 and μ_3, μ_2 respectively. The medium bounded by the curved surface is denser than both the surrounding media i.e. ($\mu_1 < \mu_2 > \mu_3$). In figure-1 the concave faces bound the denser medium.

Ray diagram: A point object O is kept on the common principal axis in the medium of refractive index μ_1. A ray from O is incident at any point Q_1 on the curved surface A_1B_1 & afterrefraction since it is going from rarer to denser medium it bends towards the normal and is refracted along Q_1Q_2. Another ray from O incident along the the principal axis OP_1 being incident normally passes without any deviation along P_1P_2, these two refracted rays Q_1Q_2 and P_1P_2 meet at I_1 directly in figure-1. For the curved surface A_2B_2, Q_1Q_2 & P_1P_2 are incident rays since they appear to converge (in figure-1) hence I_1 serves as object for A_2B_2. Since Q_1Q_2 goes from denser to rarer medium bends away from the normal where as P_1P_2 passes without any deviation & meet at I. Hence I is the image of the object O formed by the double spherical surface.

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Calculation: We know that for refraction at a single spherical surface with object in the rarer medium

$$\frac{\mu_o}{u}+\frac{\mu_i}{v}=\frac{\mu_i-\mu_o}{r}\rightarrow(1)$$

For refraction at curved surface A_1B_1
Refractive index of the object medium $\mu_o=\mu_1$
Refractive Index of the image medium $\mu_i=\mu_2$
Object distance $P_1O=u$
Image distance$=P_1I=v_1$
Radius of curvature $=r_1$
Using equation (1)

$$\frac{\mu_1}{u}+\frac{\mu_2}{v}=\frac{\mu_2-\mu_1}{r_1}\rightarrow(2)$$

For refraction at curved surface A_2B_2
Refractive index of the object medium $\mu_o=\mu_2$
Refractive index of the image medium $\mu_i=\mu_3$
Since $\mu_2>\mu_3$ therefore $\mu_o>\mu_i$, since the object is in denser medium we have to
Consider sign correction before using the equation (1).
Figure- 1 : Object distance = $P_2I_1=u'$(say)=Negative
Image distance = $P_2I=v$=Positive
Radius of curvature $=r_2$= Negative

Using equation (1)

$$\frac{\mu_2}{-u'}+\frac{\mu_3}{v}=\frac{\mu_3-\mu_2}{-r_2}$$

$$\frac{\mu_2}{-u'}+\frac{\mu_3}{v}=\frac{\mu_2-\mu_3}{r_2}\rightarrow(3)$$

adding equation (2) & (3)

$$\frac{\mu_1}{u}+\frac{\mu_2}{v_1}-\frac{\mu_2}{u'}+\frac{\mu_3}{v}=\frac{\mu_2-\mu_1}{r_1}+\frac{\mu_2-\mu_3}{r_2}$$

$$v_1=P_1I=P_2I\pm P_1P_2=u'\pm t$$

$$P_1P_2=t=\text{thickness of the lens for thin lens } v_1=u'$$

$$\frac{\mu_1}{u}+\frac{\mu_3}{v}=\frac{\mu_2-\mu_1}{r_1}+\frac{\mu_2-\mu_3}{r_2}$$

Above equation gives the relation between the object distance image distance and radius of curvature for refraction through a lens, when the two surrounding media are different.

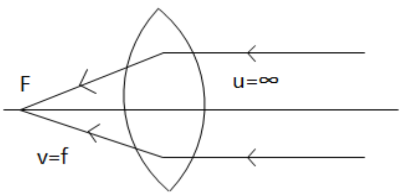
Special Case I: Let both the media surrounding the lens be same $\mu_1=\mu_3$

$$\frac{\mu_1}{u}+\frac{\mu_1}{v}=\frac{\mu_2-\mu_1}{r_1}+\frac{\mu_2-\mu_1}{r_2}$$

$$\frac{1}{u}+\frac{1}{v}=\left(\frac{\mu_2-\mu_1}{\mu_1}\right)\left(\frac{1}{r_1}+\frac{1}{r_2}\right)$$

$$\frac{1}{u}+\frac{1}{v}=\left(\frac{\mu_2}{\mu_1}-1\right)\left(\frac{1}{r_1}+\frac{1}{r_2}\right)$$

If rays are incident parallel to the principal axis (u=∞), then the point on which image is formed is said to be the FOCUS. The distance of the focus from the optical center is known as focal length (f).



From equation

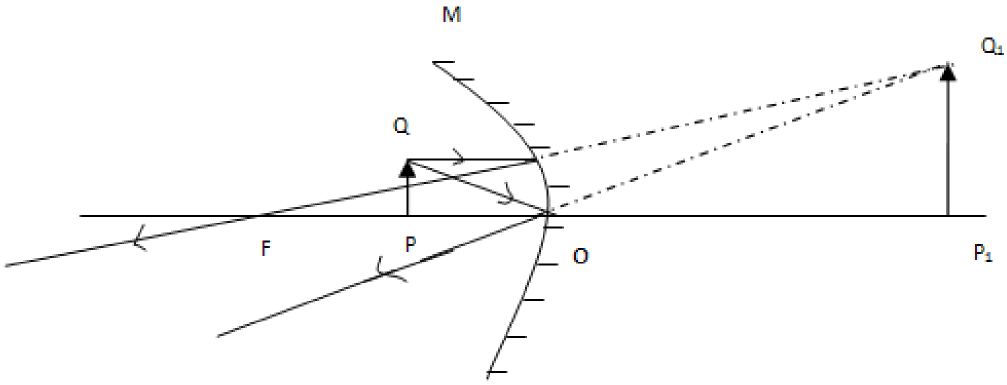
$$\frac{1}{\infty}+\frac{1}{f}=\left(\frac{\mu_2}{\mu_1}-1\right)\left(\frac{1}{r_1}+\frac{1}{r_2}\right)$$

$$\frac{1}{f}=\left(\frac{\mu_2}{\mu_1}-1\right)\left(\frac{1}{r_1}+\frac{1}{r_2}\right)$$

Equation gives the focal length of lens when the surrounding medium is other than air. Replacing µ2 by n_2 and μ_1 by n_1 ,r_1 by R_1 and R_2 by r_2 considering direction of curved surface opposite we get

$$\frac{1}{f}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

(b) Draw any ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the pole. Using this diagram, derive the magnification formula for the image formed.



PQ is the object kept between the focus F and the pole o of the concave mirror, P_1Q_1 image formed by the mirror as shown.

Magnification Produced:

ΔPOQ *and* ΔP_1OQ_1 *are similar*.

$$\frac{P_1Q_1}{PQ} = \frac{OP_1}{OP}$$

OP1= image distance = -v (as per modern sign convention image distance measured against refracted rays taken as negative)

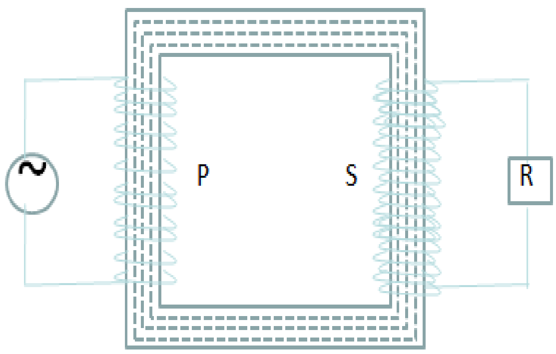
OP = object distance =+ u (as per modern sign convention object distance measured against incident rays taken as positive)

So magnification (m):

$$m = \frac{OP_1}{OP} = -\frac{v}{u}$$

Q30. (i) With the help of a labelled diagram, describe briefly the underlying principle and working of a step up transformer.

Answer:



Principle: It is a device which converts high voltage, A.C. into low voltage A.C. and vice versa. It is based upon the principle of *mutual induction*. When alternating current passed through a coil (Primary), an induced e.m.f. is set up in the neighbouring coil(Secondary).

Construction: A transformer consists of two coils of many turns of insulated copper wire wound on a closed laminated iron core. One of the coils known as primary (P) is connected to A.C. supply. The other coil known as Secondary (S) is connected to the load.

Working: It consists of two coils known as Primary (P) and Secondary(S) having different no. of turns and are would on two opposite arms of a thick laminated iron core. The alternating emf to be transformed is connected across the primary coil. The varying current flowing through the primary coil produces varying magnetic lines of force. The magnetic ion core provides an easy path for the flow of lines of force hence the lines of force flowing through the core cuts the secondary coil and induces an emf across the secondary. An induced current flow through the secondary coil and thus a potential drop is obtained across the resistance which is known as output voltage.

Theory: Given

N_p and N_s are number of turns in the primary and secondary respectively

V_p and V_s are their respective voltages.

$$\therefore V_p = -N_p \frac{d\phi}{dt} \text{ and } V_s = -N_s \frac{d\phi}{dt}$$

Where, N_p and N_s are number of turns in the primary and secondary respectively and V_p and V_s are their respective voltages.

$$\therefore \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

This ratio $\frac{N_s}{N_p}$ is called the turns ratio.

In a step − up transformer : $N_s > N_p$, *so* $V_s > V_p$.

(ii) Write any two sources of energy loss in a transformer.

Answer: Sources of energy loss in transformer

I. **Eddy currents:** The alternating magnetic flux induces eddy current in the iron core and causes heating. This effect is *reduced* by having a laminated core.

II. **Hysteresis:** This is due to repeated magnetisation and demagnetisation of the iron core. It can be reduced by using a magnetic material having low hysteresis loss.

(iii) A step up transformer converts a low input voltage into a high output voltage. Does it violate law of conservation of energy? Explain.

Answer: A transformer does not violate law conservation of energy. In step up transformer voltage increased but at the same time current reduced, in step down transformer voltage decreased but current increased, thus the product of current and voltage that is power remains same in input and output provided there is no loss.

OR

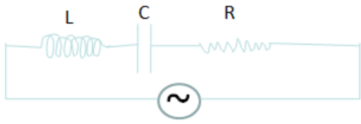
Derive an expression for the impedance of a series LCR circuit connected to an AC supply of variable frequency.

Plot a graph showing variation of current with the frequency of the applied voltage.

Explain briefly how the phenomenon of resonance in the circuit can be used in the running mechanism of a radio or a TV set.

Answer:

L-C-R circuit :



We know that in capacitor (C) current leads emf by π/2 , in inductance current lags emf by π/2 (L) and in resistance (R) current is in phase with emf.

If V_L, V_c, V_R and V represents the voltage across the inductor, capacitor, resistance and source.

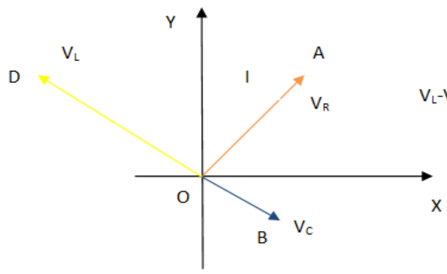


Fig-1

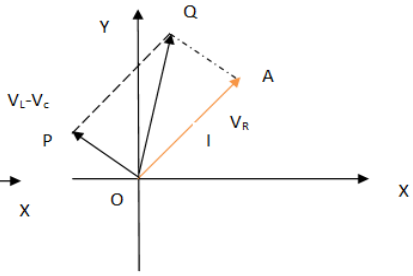


Fig-2

For the above vector diagram, Let the direction of current be along $\overrightarrow{OA}$, therefore

V_R= IR can be represented along $\overrightarrow{OA}$.

V_L=IX_L=IωL , since it lags current by π/2 , represented along $\overrightarrow{OD}$.

V_c=IXc=I/ωC, since current leads by π/2, represented along $\overrightarrow{OB}$.

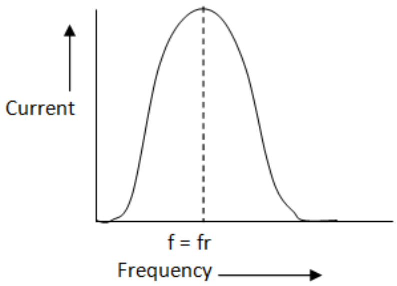
Therefore V_c and V_L are 180^0 apart, resultant obtained by subtraction= V_L- V_C=$\overrightarrow{OP}$ (shown in fig-2)

Now from ΔOQP: $OQ^2 = OP^2 + PQ^2$, $V^2= V_R^2+(V_L -V_c)^2= I^2[R^2 +(\omega L- 1/\omega C)^2]$ Or $\frac{V}{I} = Impedance(Z)$

The impedance of the circuit

$$z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

The given graph showing variation of current with the frequency of the applied voltage.



The radio and TV receiver sets are the practical applications of series resonant circuits. Signals of several different frequencies are available in air. The capacitance of the capacitor in tuning circuit is varied by turning the tuning knob of the radio/TV set. Thus the frequency of the LCR circuit varied till it matches the frequency of the derived signal.

A series resonant circuit allows maximum current through it. So, it is called acceptor circuit. A series resonant circuit is that circuit in which inductance L, capacitance C and resistance R are connected in series. The impedance of this circuit has a minimum value and the current through the circuit is maximum.

Note: Most of answers are written in descriptive way to make you comfortable for self study, however while writing in exam you need to write in brief with proper understanding to complete your answer sheet in time.